COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
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ACTIVITY NO. 9.1 **Implementing Trees**

1. PROCEDURE OUTPUT

What procedures did you perform?

Binarytree.h:

```
#ifndef BINARYTREE_H
#define BINARYTREE_H

template <typename T>
class BinaryTree{
    private:
        T key;
        // pointers for left and right child
        BinaryTree<T>* leftChild;
        BinaryTree<T>* rightChild;
    public:
        //constructor
        BinaryTree(T rootObj);
        // insert a child on the left node
        void insertChildLeft(T newNode);
        void insertChildRight(T newNode);

        BinaryTree<T>* getLeftChild();
        BinaryTree<T>* getRightChild();

        T getRootVal();
        void setRootVal(T newNode);
};

//constructor:
template<typename T>
BinaryTree<T>::BinaryTree(T rootObj){
```

```
    key = rootObj;
    leftChild = nullptr;
    rightChild = nullptr;
}

// insert child:
template<typename T>
void BinaryTree<T>::insertChildLeft(T newNode){
    if(leftChild == nullptr){
        leftChild = new BinaryTree<T>(newNode);
    }else{
        BinaryTree<T>* newNode_ = new BinaryTree<T>(newNode);
        // move the current left child below the new Node:
        newNode_->leftChild = leftChild;

        leftChild = newNode_;
    }
}

template<typename T>
void BinaryTree<T>::insertChildRight(T newNode){
    if(rightChild == nullptr){
        rightChild = new BinaryTree<T>(newNode);
    }else{
        BinaryTree<T>* newNode_ = new BinaryTree<T>(newNode);
        newNode_->rightChild = rightChild;
        rightChild = newNode_;
    }
}

template<typename T>
BinaryTree<T> *BinaryTree<T>::getLeftChild(){
    return leftChild;
}

template<typename T>
BinaryTree<T> *BinaryTree<T>::getRightChild(){
    return rightChild;
```

```
}

template<typename T>
T BinaryTree<T>::getRootVal(){
    return key;
}

template<typename T>
void BinaryTree<T>::setRootVal(T newNode){
    key = newNode;
}

#endif
```

Main.cpp:

```
#include <iostream>
#include "binaryTree.h"

main(){

    BinaryTree<char>* root = new BinaryTree<char>('A');
    root->insertChildLeft('B');
    root-> insertChildRight('C');

    root->getLeftChild()->insertChildLeft('D');
    root->getLeftChild() -> insertChildRight('E');
    root -> getRightChild()->insertChildRight('F');


    std::cout<<"The root: " << root->getRootVal()<< std::endl;
    std::cout<<"The left child of root A is: "
        << root->getLeftChild()->getRootVal() <<std::endl;
    std::cout<<"The left child of node B is:: "
        << root->getLeftChild()->getLeftChild()->getRootVal()
<<std::endl;
    std::cout<<"The right child of node B is: : "
        << root->getLeftChild()->getRightChild()->getRootVal()
<<std::endl;
```

```
    std::cout<<"The right child of root A is: "
        << root->getRightChild()->getRootVal() <<std::endl;
    std::cout<<"The right child of node C is:: "
        << root->getRightChild()->getRightChild()->getRootVal()
<<std::endl;

    root -> getRightChild()->setRootVal('V');
    std::cout<<"The NEW right child of root A is: "
        << root->getRightChild()->getRootVal() <<std::endl;


    /// test it here:




    return 0;
}
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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